ABDULLA JASEM ALMANSOORI

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RESEARCH SUMMARY

My research focuses on efficient collaborative learning via low-rank adaptation. I develop and analyze methods for personalized federated learning, low-rank adaptation, and distributed optimization under privacy, heterogeneity, and communication constraints.

Research Interests: Personalized federated learning, low-rank adaptation, efficient optimization for large models, variational learning, healthcare AI.

EDUCATION

Mohamed bin Zayed University of AI (MBZUAI) — Abu Dhabi, UAE
PhD in Machine Learning (Advisors: Martin Takáč & Samuel Horváth) — (Aug 2021 - May 2026)
Thesis: Adapters for Collaborative Learning: Personalization and Structure-Aware Optimization

University of Southern California (USC) — Los Angeles, CA
MS in Computer Science — (Jan 2017 - Dec 2018)

Purdue University — West Lafayette, IN
BS in Industrial Engineering (with distinction) — (Aug 2012 - May 2016)

WORK EXPERIENCE

RIKEN – Center for Advanced Intelligence Project (AIP) — Tokyo, Japan
Research Intern (PI: Emtiyaz Khan & Thomas Möllenhoff) — (Aug 2025 - Feb 2026)

- Worked in the Adaptive Bayesian Intelligence team. Developed schedule-free variational Newton-style methods for Bayesian deep learning, improving convergence and robustness relative to IVON baselines.

Meta – Monetization: Ranking and AI Foundations — Sunnyvale, CA
ML Engineer Intern (Mentor: Fan Xia) — (May 2025 - Aug 2025)

- Built 20M+ sample training pipelines for ads ranking, trained 7B-parameter multimodal models, and distilled 1B-parameter models for production use, improving offline AUC vs. baseline.

Sheikh Shakhbout Medical City (SSMC) — Abu Dhabi, UAE
Trainee (part-time) (Mentor: Siddiq Anwar) — (Jan 2025 - Present)

- Leading a multi-hospital federated learning study for 30-day readmission prediction in the UAE. Coordinated clinical collaboration, IRB approval, data readiness, and model development.

UAE National Service, Dubai Police — (Feb 2021 - Jun 2022)

- Finished army boot camp. Worked as a security guard in Dubai International Airport and Al Maktoum International Airport and as a vehicle inspector in Expo 2020.

New York University – Abu Dhabi (NYUAD) — Abu Dhabi, UAE
Research Assistant (PI: Kris Gunsalus) — (Jan 2020 - Feb 2021)

- Worked in the Center of Genomics and Systems Biology with the core bioinformatics team. Supported next-generation sequencing workflows and AI-based bioinformatics pipelines for genomics analysis.

PUBLICATIONS

Preprint

1. *Beyond SGD, Without SVD: Proximal Subspace Iteration LoRA with Diagonal Fractional K-FAC.* (2026)
   Abdulla Jasem Almansoori, Maria Ivanova, Andrey Veprikov, Aleksandr Beznosikov, Samuel Horváth, and Martin Takáč. arXiv preprint.

Conference

1. *Collaborative and Efficient Personalization with Mixtures of Adaptors.* (2025)
   Abdulla Jasem Almansoori, Samuel Horváth, and Martin Takáč. Conference on Parsimony and Learning (CPAL).
2. *Byzantine-Tolerant Methods for Distributed Variational Inequalities.* (2023)
   Nazarii Tupitsa, **Abdulla Jasem Almansoori**, Yanlin Wu, Martin Takáč, Karthik Nandakumar, Samuel Horváth, Eduard Gorbunov. Neural Information Processing Systems (NeurIPS).

Journal

1. *PaDPaF: Partial Disentanglement with Partially-Federated GANs.* (2024)
   Abdulla Jasem Almansoori, Samuel Horváth, and Martin Takáč. Transactions on Machine Learning Research (TMLR).
2. *Stochastic Gradient Methods with Preconditioned Updates.* (2024)
   Abdurakhmon Sadiev, Aleksandr Beznosikov, **Abdulla Jasem Almansoori**, Dmitry Kamzolov, Rachael Tappenden, and Martin Takáč. Journal of Optimization Theory and Applications (JOTA).

Workshop

1. *Faster Than SVD, Smarter Than SGD: The OPLoRA Alternating Update.* (2025)
   Abdulla Jasem Almansoori, Maria Ivanova, Andrey Veprikov, Aleksandr Beznosikov, Samuel Horváth, and Martin Takáč. OPT 2025 Workshop.
2. *PaDPaF: Partial Disentanglement with Partially-Federated GANs.* (2023)
   Abdulla Jasem Almansoori, Samuel Horváth, and Martin Takáč. FLSys 2023 Workshop (oral).

POSTERS & TALKS

1. *Schedule-Free Variational Newton Methods.* (Feb 2026)
   Talk, internal seminar, RIKEN-AIP.
2. *Collaborative and Efficient Personalization with Mixtures of Adaptors.* (Nov 2024)
   Poster, ADIA Lab Symposium 2024.
3. *Federated Personalization with Mixtures of LoRAs.* (Dec 2023)
   Poster, 2nd Collaborative Learning Workshop, MBZUAI.
4. *PaDPaF: Partial Disentanglement with Partially-Federated GANs.* (Jun 2023)
   Oral, Federated Learning Systems (FLSys) Workshop, MLSys 2023.
5. *Lower Bounds for Non-Convex Stochastic Optimization.* (Oct 2022)
   Talk, optimization seminar, MBZUAI.

ACADEMIC SERVICE

- **Teaching Assistant, MBZUAI**: (i) ML701: Machine Learning (Fall 23, Fall 24), (ii) ML802: Advanced Topics in Machine Learning (Spring 24), (iii) ML814: Selected Topics in Machine Learning (Spring 25), (iv) ML8509: Collaborative Learning (Spring 26).
- **Conference Reviewer**: (i) NeurIPS 2022-2026, (ii) ICML 2023-2026, (iii) ICLR 2023-2025, (iv) ICOMP 2024. **Workshop Reviewer**: AdaptFM@ICML 2026. **Journal Reviewer**: TMLR.

AWARDS & HONORS

- **Department of Machine Learning PhD Award**, MBZUAI, 2026.
- **ICML Gold Reviewer Award**, 2026.
- **ICML Best Reviewer Award**, 2024.
- **MBZUAI Executive Program** Project Coordinator, 4th Cohort (2023).
  - Coordinated an AI strategy project for the government with executive-level participants.
- **Kawader Research Assistantship**, NYUAD, 2020.
- **Phi Beta Kappa** Society, Purdue University, 2016.
- **Distinguished Students Scholarship**, UAE Scholarships Office (SCO), 2012.

SKILLS

Programming Languages	Python, C, JS, Haskell
ML/Research	PyTorch, Hugging Face, federated learning (flwr)
Experimentation/Systems	Hydra, Slurm, Docker, Git
Languages	Arabic (native), English (fluent), Japanese (intermediate), Italian (intermediate)